

M11 Module Review

Storage & File Formats Foundations | post-attempt review

Open only after genuine attempt

Use this to diagnose reasoning, not to replace the first attempt.

Review L01 - Row vs columnar storage; where Parquet fits

Expected reasoning: You can distinguish file format from table/database semantics.

Repair trigger: Calling Parquet a database or assuming it provides transactions by itself.

Review L02 - Parquet schema, row groups and metadata

Expected reasoning: You can explain metadata versus data pages.

Repair trigger: Memorizing one writer's row-group count as universal.

Review L03 - Projection, predicates and analytical reads

Expected reasoning: You can describe what was requested and what evidence the engine exposes.

Repair trigger: Saying predicate pushdown always eliminates all unnecessary I/O.

Review L04 - Compression, small files and compaction

Expected reasoning: You can explain small-file cost without inventing a magic target size.

Repair trigger: Compacting without reconciliation or assuming one giant file is always best.

Review L05 - Partitioning strategy and over-partitioning

Expected reasoning: You can state query pattern, cardinality and size reasoning.

Repair trigger: Partitioning by customer_id only because it is an important business key.

Review L06 - Lake vs warehouse vs lakehouse

Expected reasoning: You can distinguish storage layer from table-management layer.

Repair trigger: Assuming a folder of Parquet files is automatically a trustworthy lakehouse table.

Review L07 - Delta tables, transaction log and ACID concepts

Expected reasoning: You can explain what makes a Delta table more than Parquet files.

Repair trigger: Deleting log files or manually swapping Parquet files to simulate a transaction.

Review L08 - MERGE/upsert and current-state records

Expected reasoning: You can defend key, update rule and rerun behavior.

Repair trigger: MERGE without a stable key or version rule, causing duplicates/incorrect winners.

Review L09 - Schema enforcement, controlled evolution and compatibility

Expected reasoning: You can state why the change is safe or breaking.

Repair trigger: Automatically accepting every new column/type because the writer can.

Review L10 - Time travel, retention, VACUUM safety and medallion checkpoint

Expected reasoning: You can defend file/table format, layer and recovery choices together.

Repair trigger: Running VACUUM before required recovery evidence or mixing business aggregates into Bronze.